a2-code-540840166-541002619

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1 COMP4318/5318 Assignment 2: Image Classification

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1.1 Setup and dependencies

[ ]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

import random as rd
from collections import Counter
from sklearn.decomposition import PCA
from sklearn.model_selection import StratifiedKFold, train_test_split,␣
↪ParameterGrid, cross_val_score
from sklearn.metrics import accuracy_score
from sklearn.ensemble import RandomForestClassifier
import tensorflow as tf
from tensorflow import keras
from sklearn.preprocessing import StandardScaler
import time
from datetime import datetime
from sklearn.metrics import f1_score
import keras_tuner
```

[2]:
```
# Ignore future warnings
from warnings import simplefilter
simplefilter(action='ignore', category=FutureWarning)
```

[3]:
```
np.random.seed(0)
rd.seed(0)
```

[4]:
```
## Setting the 10 fold stratified cross-validation
cvKFold=StratifiedKFold(n_splits=10, shuffle=True, random_state=0)
```

1.2 1. Data loading, exploration, and preprocessing

Load the dataset.

[5]:
```
# Load the dataset training and test sets as numpy arrays
# assuming Assignment2Data folder is present in the same directory
# as the notebook
X_train = np.load('Assignment2Data/X_train.npy')
y_train = np.load('Assignment2Data/y_train.npy')
X_test = np.load('Assignment2Data/X_test.npy')
y_test = np.load('Assignment2Data/y_test.npy')
```

To find the number of samples in each set, we can examine the shape of the data arrays.

[6]:
```
# Check the shape of each set.
print(f"Shape of X_train: {X_train.shape}")
print(f"Shape of y_train: {y_train.shape}")
print(f"Shape of X_test: {X_test.shape}")
print(f"Shape of y_test: {y_test.shape}")
```

```
Shape of X_train: (32000, 28, 28, 3)
Shape of y_train: (32000,)
Shape of X_test: (8000, 28, 28, 3)
Shape of y_test: (8000,)
```

We have 32000 and 8000 figures in train and test dataset respectively. Each figure is generated by 28*28 pixels. Each pixel contains three values corresponding to RGB.

Different from the tutorial, now we have three channels as input value rather than only one channel containing black or white.

[7]:
```
# Prepare the data for imshow.

X_train_imshow = X_train
```

Convert the value in the data to the float. Because large inputs can cause gradients very close to 0 after passing the neuron output through an activation function. So we need to convert all value from 0~255 to 0~1. This can be done simply through dividing by 255.

[8]:
```
# A Normalization.
def data_process_Normalization(x):
    x = x / 255
    return x

X_train = data_process_Normalization(X_train)
X_test = data_process_Normalization(X_test)
```

[9]:
```
X_train.shape
```

[9]: (32000, 28, 28, 3)

[10]:
```
# StandardScaler
def pre_process_StandardScaler(X_train,X_test):
```

```
    X_train_shape = X_train.shape
    X_test_shape = X_test.shape
    X_train = X_train.reshape(X_train.shape[0],-1) # flat the x.
    X_test = X_test.reshape(X_test.shape[0],-1)

    scaler = StandardScaler()
    scaler.fit(X_train)
    X_scaled_train = scaler.transform(X_train)
    X_scaled_test = scaler.transform(X_test)

    res_train = X_scaled_train.reshape(X_train_shape)
    res_test = X_scaled_test.reshape(X_test_shape)

    return res_train,res_test

X_train,X_test = pre_process_StandardScaler(X_train,X_test)
```

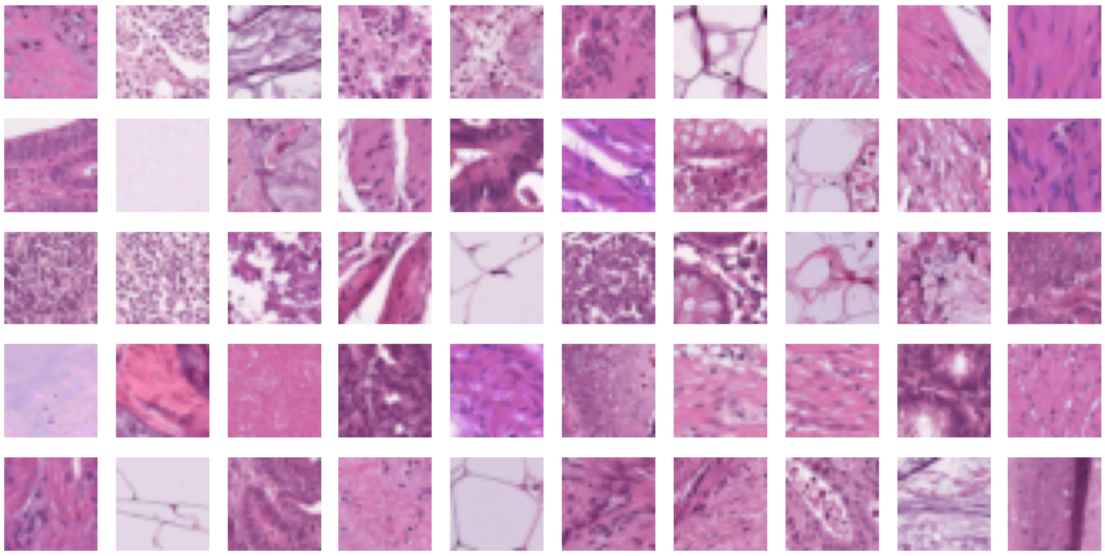
[11]:
```
# Function to flat the dimension.

def flat_dimension(x):

    x = x.reshape(x.shape[0],-1)

    return x

flat_X_train = flat_dimension(X_train)
flat_X_test = flat_dimension(X_test)
```

1.2.1 Examples of preprocessed data

Let us see some examples in training dataset.

[12]:
```
# A function to show the examples in trainging dataset.

def examples_all(df, n_rows = 5, n_cols = 10):

    plt.figure(figsize=(n_cols, n_rows))
    plt.axis('off')

    for r in range(n_rows):
        for c in range(n_cols):

            index = r * 10 + c

            plt.subplot(n_rows, n_cols, index+1)
            plt.axis('off')
```

```
            plt.imshow(df[index], cmap= "binary" )


    plt.show()

examples_all(X_train_imshow)
print(f"First 10 examples of y_train: {y_train[0:10]}")
```



```
First 10 examples of y_train: [7 3 4 2 6 2 0 7 5 5]
```

From the background knowledge, the PathMNIST is a medical image classification dataset derived from multiple tissue slice images.

The PathMNIST has 9 types of tissues as outputs

Let us check if the class in y is 0~8.

[13]:
```
print(f"The class in y_train : {np.unique(y_train)}")
# The ans is yes.
```

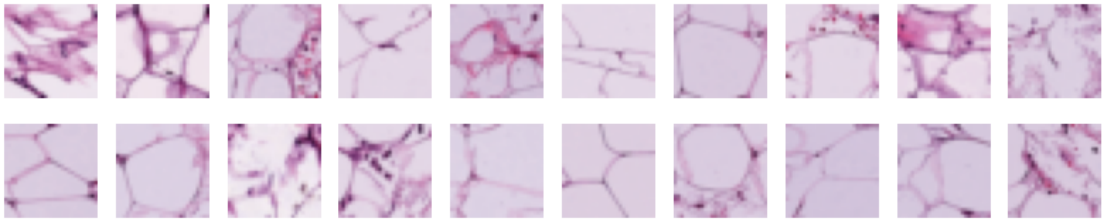
```
The class in y_train : [0 1 2 3 4 5 6 7 8]
```

I would like to see some examples of different tissues.

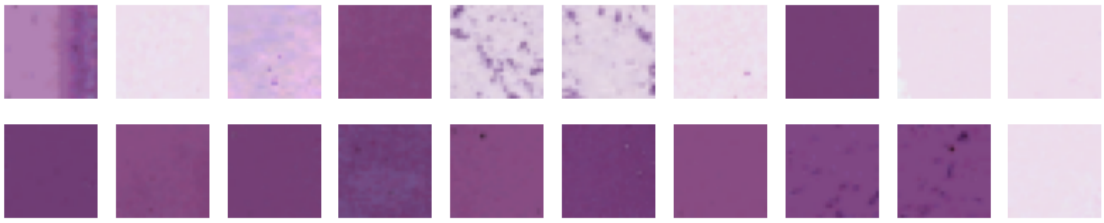
[14]:
```
def example_single(y_value, n_cols = 10, n_rows = 2):

    plt.figure(figsize=(n_cols, n_rows))
    plt.axis('off')

    total = n_cols * n_rows
    i = 1
```

```
    index = 0
    index_list = []

    while i <= total:
        if y_train[index] == y_value:
            index_list.append(index)
            index += 1
            i += 1
        else:
            index += 1

    for r in range(n_rows):
        for c in range(n_cols):

            index = r * 10 + c

            plt.subplot(n_rows, n_cols, index+1)
            plt.axis('off')
            plt.imshow(X_train_imshow[index_list[index-1]], cmap= "binary" )

    plt.show()

def example_single_print():
    for i in range(9):
        print(f"The example figure of 'y = {i}' type.")
        example_single(i)

example_single_print()
```

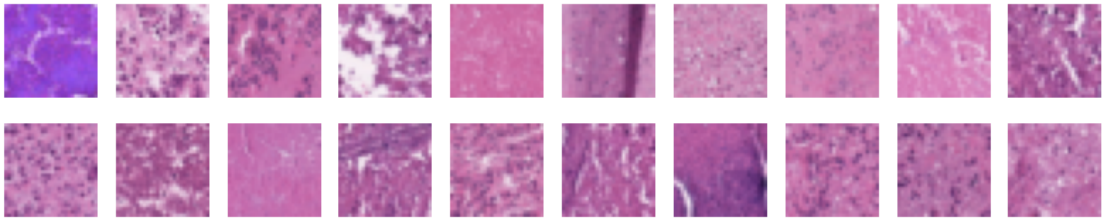
```
The example figure of 'y = 0' type.
```



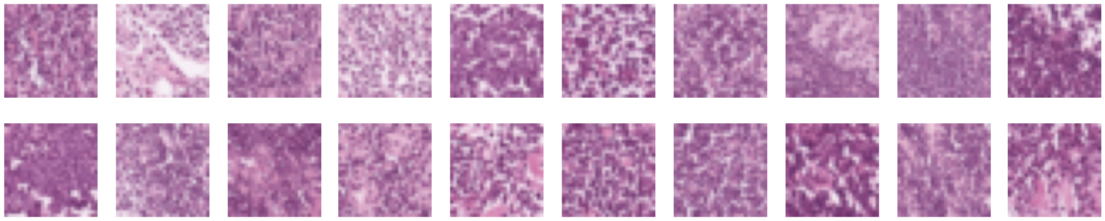
```
The example figure of 'y = 1' type.
```



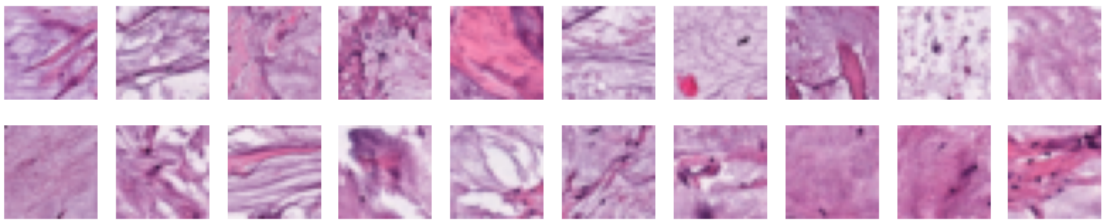
The example figure of 'y = 2' type.



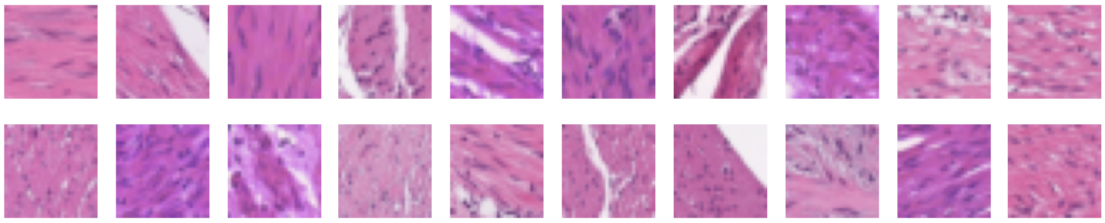
The example figure of 'y = 3' type.



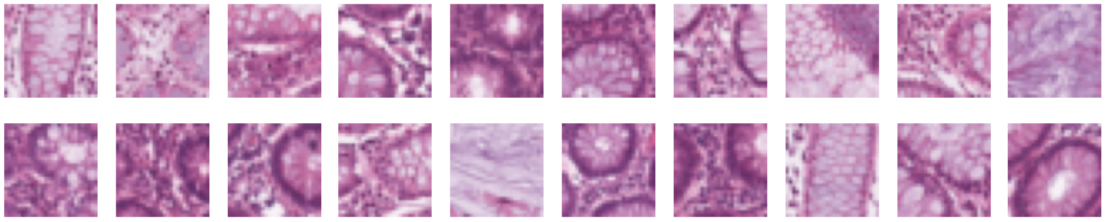
The example figure of 'y = 4' type.



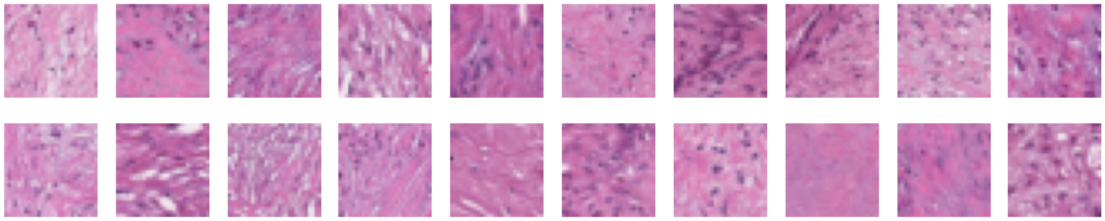
The example figure of 'y = 5' type.



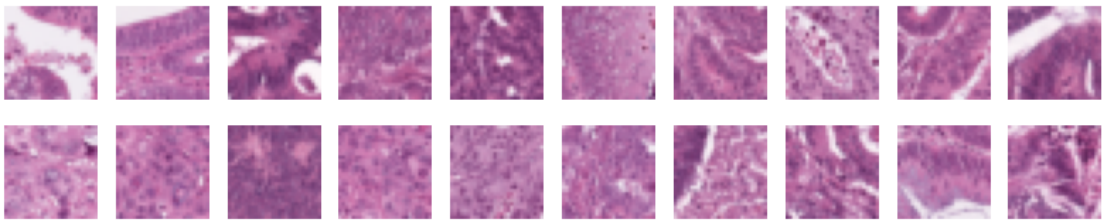
```
The example figure of 'y = 6' type.
```



```
The example figure of 'y = 7' type.
```



```
The example figure of 'y = 8' type.
```



Here I also want to see whether the tissue images of each type are evenly distributed in the training set and the test set.

[15]:
```
# A function to check the percentage of tissue type.

def tissue_type_rate():

    train = Counter(y_train)
    test = Counter(y_test)
    y_unique = np.unique(y_train)

    res_train = pd.DataFrame()
    res_test = pd.DataFrame()

    for i in y_unique:
        df_train = pd.DataFrame([train[i],train[i]/32000]).T
        res_train = pd.concat([res_train, df_train])

        df_test = pd.DataFrame([test[i],test[i]/8000]).T
        res_test = pd.concat([res_test, df_test])

    res_train.index = (y_unique)
    res_train.columns = ['Quantity', 'Rate']

    res_test.index = (y_unique)
    res_test.columns = ['Quantity', 'Rate']

    print('y distubeted in training set:\n {}'.format(res_train))
    print('y distubeted in testing set:\n {}'.format(res_test))

tissue_type_rate()
```

```
y distubeted in training set:
    Quantity      Rate
0    3490.0  0.109063
1    3431.0  0.107219
2    3505.0  0.109531
3    3656.0  0.114250
4    2950.0  0.092188
5    4290.0  0.134063
6    2728.0  0.085250
7    3253.0  0.101656
8    4697.0  0.146781
y distubeted in testing set:
    Quantity      Rate
0     873.0  0.109125
1     858.0  0.107250
2     877.0  0.109625
3     914.0  0.114250
4     737.0  0.092125
5    1072.0  0.134000
```

```
6     682.0  0.085250
7     813.0  0.101625
8    1174.0  0.146750
```

So the types of y are evenly disbuted in training and testing set. Even through, there are still existed tiny difference between differnet tissues. When reducing training size, I still need to follow the 'stratify = y'.

1.3 2. Algorithm design and setup

1.3.1 Algorithm of choice from first six weeks of course

Normal machine learning model need two dimensions as input vector. So before input the data, we need to decrease the dimensions from four to two.

For some simple algorithm, they are not suit for the image regonition task.

Here I only select two algorithm result to make sure they are not suitable.

[16]:
```
# PCA method to decrease the deminsion.
def PCA_train(train, test):
    pca = PCA(n_components=0.95)  # Keep 95% information.

    pca_fit = pca.fit(train)

    res_train = pca_fit.transform(train)
    res_test = pca_fit.transform(test)
    print(f"After PCA method, remain dimensions:{res_train.shape[1]}")

    return res_train,res_test
```

[17]:
```
flat_X_train_PCA,flat_X_test_PCA = PCA_train(flat_X_train,flat_X_test)
```

```
After PCA method, remain dimensions:343
```

[18]:
```
# RandonForest algorithm setup.

def rf(n_estimators = 10,
       max_leaf_nodes = 10,
       X_test = None,
       y_test = None
       ):

   start = time.time()

   rf = RandomForestClassifier(n_estimators = n_estimators, max_leaf_nodes =␣
 ↪max_leaf_nodes,
                               criterion='entropy', max_features = 'sqrt',␣
 ↪random_state=0)
```

```
    X_train_rf, X_valid_rf, y_train_rf, y_valid_rf =␣
↪train_test_split(flat_X_train_PCA, y_train, train_size=0.9, random_state=0)

    if (X_test is None) and (y_test is None):
        X_test, y_test = X_valid_rf, y_valid_rf

    rf.fit(X_train_rf,y_train_rf)
    y_pred = rf.predict(X_test)

    end = time.time()

    Duration = round(end-start,3)
    acc_rf = round(accuracy_score(y_test, y_pred),3)
    f1_m = round(f1_score(y_test, y_pred, average='macro'),3)
    f1_w = round(f1_score(y_test, y_pred, average='weighted'),3)

    print("Rf Accuracy: {:.3f}".format(acc_rf))
    print("Rf Duration(s): {:.3f}".format(Duration))
    print("Rf F1 Macro: {:.3f}".format(f1_m))
    print("Rf F1 Weighted(s): {:.3f}".format(f1_w))

    return acc_rf,Duration,f1_m,f1_w
```

[19]:
```
rf()
```

```
Rf Accuracy: 0.395
Rf Duration(s): 2.313
Rf F1 Macro: 0.296
Rf F1 Weighted(s): 0.310
```

[19]: (0.395, 2.313, 0.296, 0.31)

The RandomForest algorithm give out a accuracy with 42.9%.

Although it is not good enough, I still choose RandomForest to start tuning hyperparameter.

1.3.2 Fully connected neural network

[20]:
```
# Function to train MLP model.

def MLP(Dense_1 = 100,
        Activation_1 = 'relu',
        Dense_2 = 100,
        Activation_2 = 'relu',
        Learning_rate = 0.1,
        X_test = None,
        y_test = None
        ):
    # Ensure stability across runs
```

```
keras.backend.clear_session()
tf.random.set_seed(0)

start = time.time()

# Define our MLP layer by layer
model = keras.models.Sequential([
    keras.layers.Input([28, 28, 3]),
    keras.layers.Flatten(),
    keras.layers.Dense(Dense_1, activation = Activation_1),
    keras.layers.Dense(Dense_2, activation = Activation_2),
    keras.layers.Dense(9, activation = "softmax")
    ])

model.summary()

# Instantiate optimiser and compile the model.
opt = keras.optimizers.Adam(learning_rate=Learning_rate)
model.compile(loss='sparse_categorical_crossentropy',
            optimizer=opt,
            metrics=['accuracy'])

# Train the classifier.

X_train_MLP, X_valid_MLP, y_train_MLP, y_valid_MLP =␣
↪train_test_split(X_train, y_train, train_size=0.9, random_state=0)

if (X_test is None) and (y_test is None):
    X_test, y_test = X_valid_MLP, y_valid_MLP

history = model.fit(X_train_MLP, y_train_MLP, epochs=10, batch_size=128,
                    validation_data=(X_test, y_test))

end = time.time()

# Convert the history dictionary to a Pandas dataframe and extract the␣
↪accuracies
accuracies = pd.DataFrame(history.history)[['accuracy', 'val_accuracy']]
losses = pd.DataFrame(history.history)[['loss', 'val_loss']]

# Plot the accuracies
accuracies.plot(figsize=(8, 5))
plt.grid(True)
plt.gca().set_ylim(0, 1)
plt.xlabel('Epoch')
plt.show()
```

```
    # Plot the losses
    losses.plot(figsize=(8, 5))
    plt.grid(True)
    plt.gca().set_ylim(0, 5)
    plt.xlabel('Epoch')
    plt.show()

    # Predict
    y_pred_probs = model.predict(X_test)
    y_pred = np.argmax(y_pred_probs, axis=1) # Label the output
    f1_m = round(f1_score(y_test, y_pred, average='macro'),3)

    acc_MLP = round(accuracies['val_accuracy'][9],3)
    loss_MLP = round(losses['val_loss'][9],3)
    Duration = round(end-start,3)

    print(f'MLP Final Accuracy :{acc_MLP}')
    print(f'MLP Duration(s) :{Duration}')
    print(f'MLP Final Loss :{loss_MLP}')
    print(f'MLP Final F1(Macro) :{f1_m}')


    return acc_MLP, Duration, loss_MLP, f1_m
```

[21]:
```
MLP()
```

Model: "sequential"

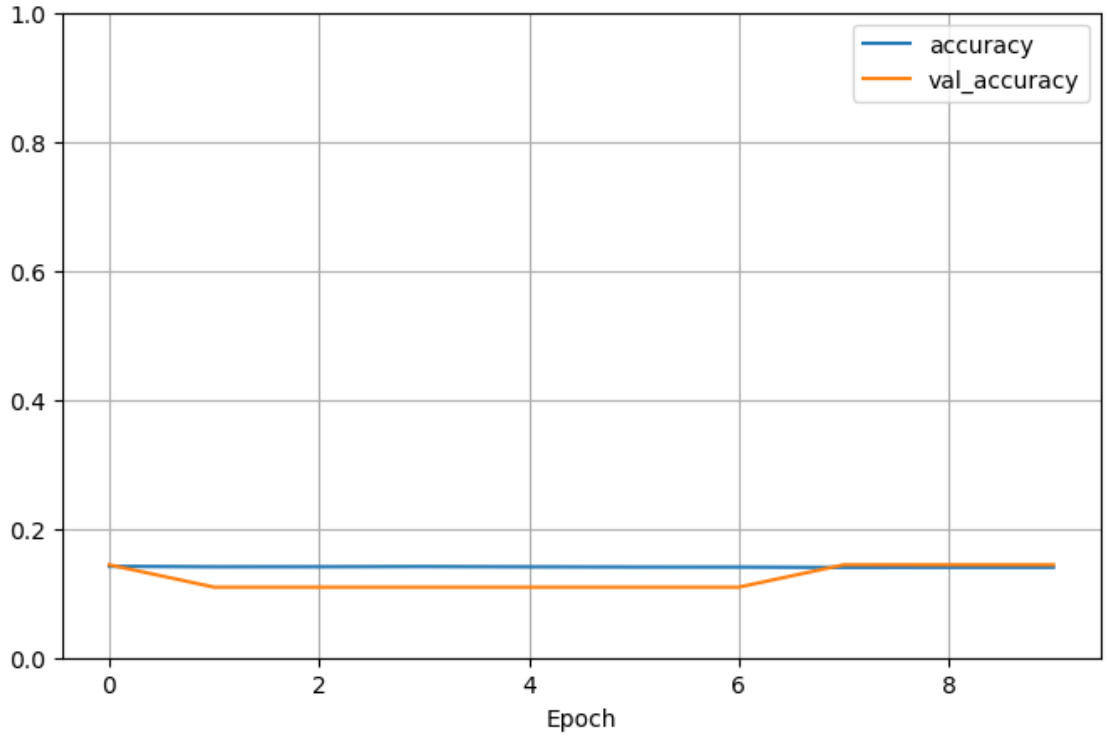
Layer (type)	Output Shape	Param #
flatten (Flatten)	(None, 2352)	0
dense (Dense)	(None, 100)	235,300
dense_1 (Dense)	(None, 100)	10,100
dense_2 (Dense)	(None, 9)	909

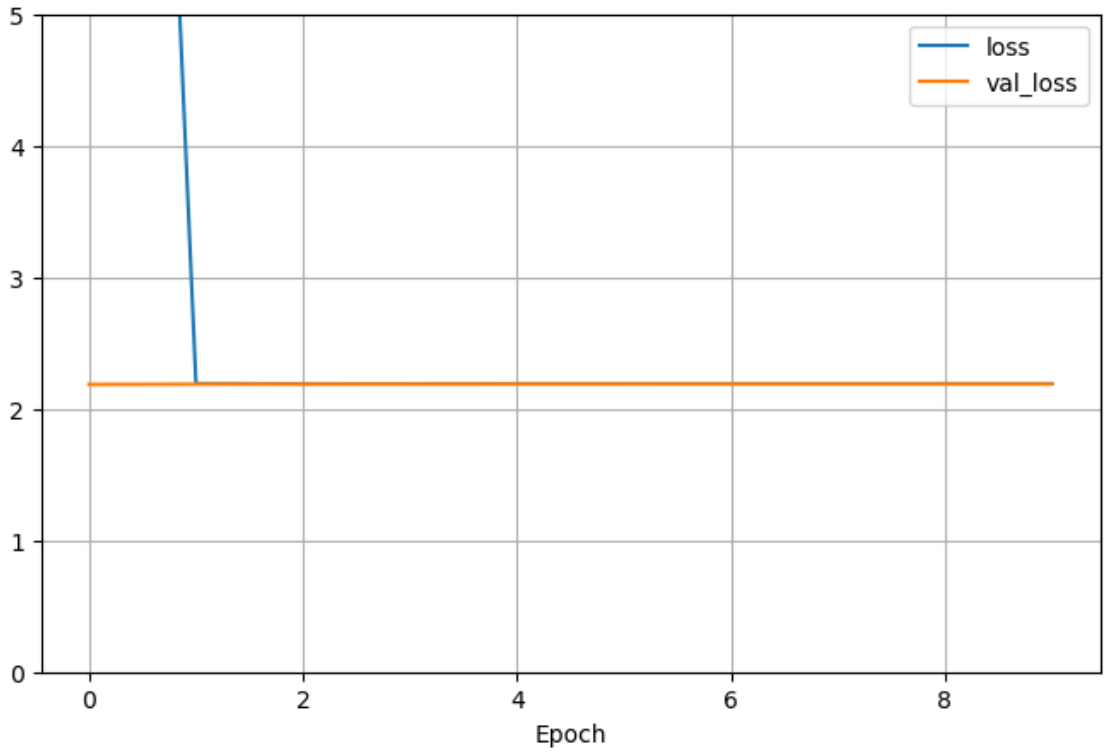
Total params: 246,309 (962.14 KB)

Trainable params: 246,309 (962.14 KB)

Non-trainable params: 0 (0.00 B)

```
Epoch 1/10
225/225              1s 2ms/step -
accuracy: 0.1376 - loss: 82.2460 - val_accuracy: 0.1441 - val_loss: 2.1857
Epoch 2/10
225/225              0s 1ms/step -
accuracy: 0.1410 - loss: 2.1910 - val_accuracy: 0.1094 - val_loss: 2.1875
Epoch 3/10
225/225              0s 1ms/step -
accuracy: 0.1406 - loss: 2.1887 - val_accuracy: 0.1094 - val_loss: 2.1881
Epoch 4/10
225/225              0s 1ms/step -
accuracy: 0.1409 - loss: 2.1891 - val_accuracy: 0.1094 - val_loss: 2.1885
Epoch 5/10
225/225              0s 1ms/step -
accuracy: 0.1404 - loss: 2.1894 - val_accuracy: 0.1094 - val_loss: 2.1886
Epoch 6/10
225/225              0s 1ms/step -
accuracy: 0.1401 - loss: 2.1895 - val_accuracy: 0.1094 - val_loss: 2.1887
Epoch 7/10
225/225              0s 1ms/step -
accuracy: 0.1401 - loss: 2.1897 - val_accuracy: 0.1094 - val_loss: 2.1888
Epoch 8/10
225/225              0s 2ms/step -
accuracy: 0.1391 - loss: 2.1898 - val_accuracy: 0.1441 - val_loss: 2.1888
Epoch 9/10
225/225              0s 1ms/step -
accuracy: 0.1397 - loss: 2.1898 - val_accuracy: 0.1441 - val_loss: 2.1888
Epoch 10/10
225/225              0s 1ms/step -
accuracy: 0.1397 - loss: 2.1899 - val_accuracy: 0.1441 - val_loss: 2.1888
```


1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
Epoch
accuracy
val_accuracy


5
4
3
2
1
0
0
2
4
6
8
Epoch
loss
val_loss

```
100/100              0s 377us/step
MLP Final Accuracy :0.144
MLP Duration(s) :3.657
MLP Final Loss :2.189
MLP Final F1(Macro) :0.028
```

[21]: (0.144, 3.657, 2.189, 0.028)

1.3.3 Convolutional neural network

[22]:
```
# Function to use Convolutional neural network.

def CNN(Activation_1 = 'relu',
        Activation_2 = 'relu',
        Dropout_rate = 0.2,
        Learning_rate = 0.1,
        X_test = None,
        y_test = None
        ):

    start = time.time()

    # Create validation set from the training set
    X_train_CNN, X_valid_CNN, y_train_CNN, y_valid_CNN =␣
 ↪train_test_split(X_train, y_train, train_size=0.9, random_state=0)

    model = keras.Sequential([

    # Specify the input shape
    keras.Input(shape=(28, 28, 3)),

    # Conv and pool block 1
    keras.layers.Conv2D(32, kernel_size=(3, 3), activation=Activation_1),
    keras.layers.MaxPooling2D(pool_size=(2, 2)),

    # Conv and pool block 2
    keras.layers.Conv2D(64, kernel_size=(3, 3), activation=Activation_2),
    keras.layers.MaxPooling2D(pool_size=(2, 2)),

    # Flatten and classify using dense output layer
    keras.layers.Flatten(),
    keras.layers.Dropout(Dropout_rate),
    keras.layers.Dense(9, activation="softmax"),
    ])

    model.summary()
```

```
# Compile model with loss, optimiser, and metrics
opt = keras.optimizers.Adam(learning_rate=Learning_rate)
model.compile(loss='sparse_categorical_crossentropy',
            optimizer=opt,
            metrics=['accuracy'])

if (X_test is None) and (y_test is None):
    X_test, y_test = X_valid_CNN, y_valid_CNN

history = model.fit(X_train_CNN, y_train_CNN, batch_size=128,
                    epochs=10, validation_data=(X_test, y_test))

end = time.time()
Duration = round(end-start,3)

# Convert the history dictionary to a Pandas dataframe and extract the␣
↪accuracies
accuracies = pd.DataFrame(history.history)[['accuracy', 'val_accuracy']]
losses = pd.DataFrame(history.history)[['loss', 'val_loss']]

# Plot the accuracies
accuracies.plot(figsize=(8, 5))
plt.grid(True)
plt.gca().set_ylim(0, 1)
plt.xlabel('Epoch')
plt.show()

# Plot the losses
losses.plot(figsize=(8, 5))
plt.grid(True)
plt.gca().set_ylim(0, 5)
plt.xlabel('Epoch')
plt.show()

# Predict
y_pred_probs = model.predict(X_test)
y_pred = np.argmax(y_pred_probs, axis=1) # Label the output
f1_m = round(f1_score(y_test, y_pred, average='macro'),3)

acc_CNN = round(accuracies['val_accuracy'][9],3)
loss_CNN = round(losses['val_loss'][9],3)

print(f'CNN Final Accuracy :{acc_CNN}')
print(f'CNN Duration(s) :{Duration}')
print(f'CNN Final loss :{loss_CNN}')
print(f'CNN Final F1(Macro) :{f1_m}')
```

```
    return acc_CNN, Duration, loss_CNN, f1_m
```

[23]:
```
CNN()
```

Model: "sequential_1"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 26, 26, 32)	896
max_pooling2d (MaxPooling2D)	(None, 13, 13, 32)	0
conv2d_1 (Conv2D)	(None, 11, 11, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 5, 5, 64)	0
flatten_1 (Flatten)	(None, 1600)	0
dropout (Dropout)	(None, 1600)	0
dense_3 (Dense)	(None, 9)	14,409

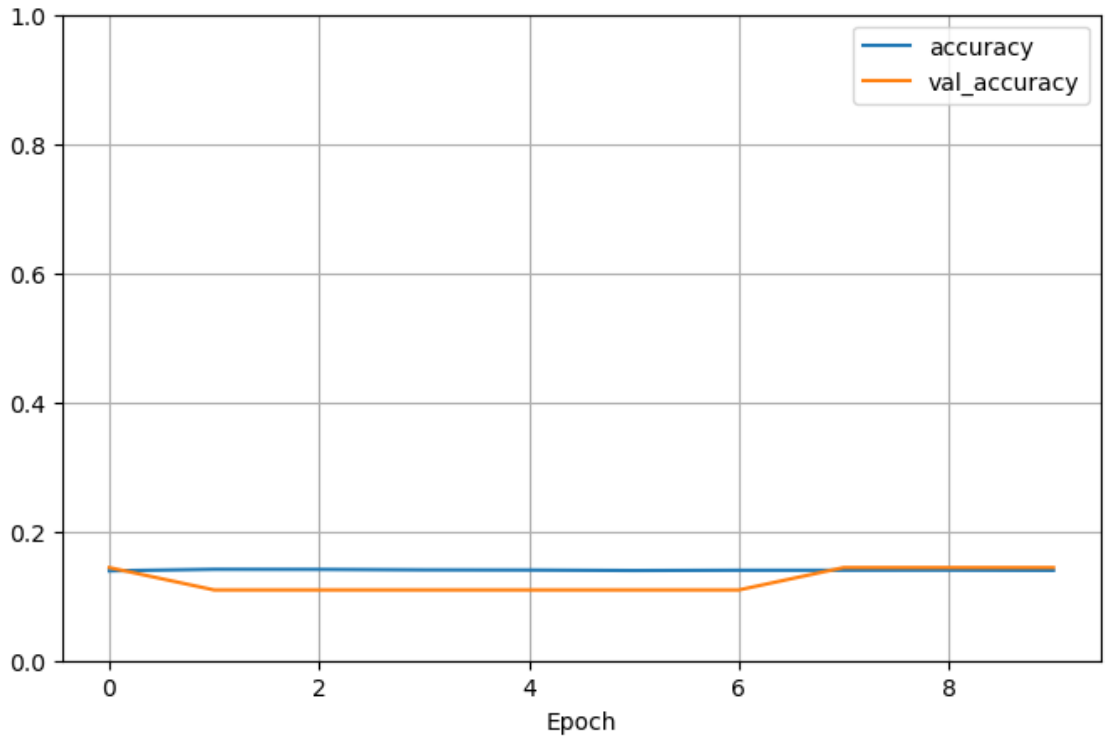
Total params: 33,801 (132.04 KB)

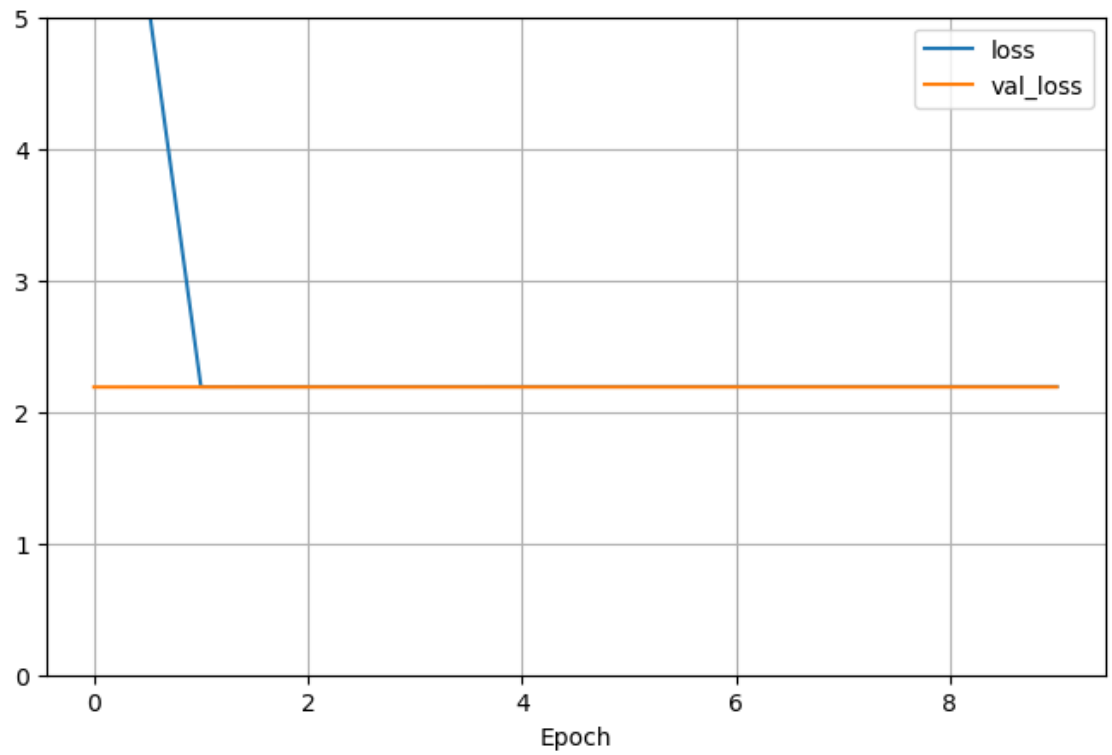
Trainable params: 33,801 (132.04 KB)

Non-trainable params: 0 (0.00 B)

```
Epoch 1/10
225/225              3s 14ms/step -
accuracy: 0.1310 - loss: 29.1900 - val_accuracy: 0.1441 - val_loss: 2.1886
Epoch 2/10
225/225              3s 14ms/step -
accuracy: 0.1409 - loss: 2.1890 - val_accuracy: 0.1094 - val_loss: 2.1889
Epoch 3/10
225/225              3s 15ms/step -
accuracy: 0.1407 - loss: 2.1894 - val_accuracy: 0.1094 - val_loss: 2.1888
Epoch 4/10
225/225              3s 15ms/step -
accuracy: 0.1399 - loss: 2.1896 - val_accuracy: 0.1094 - val_loss: 2.1888
Epoch 5/10
225/225              3s 15ms/step -
```

```
accuracy: 0.1398 - loss: 2.1897 - val_accuracy: 0.1094 - val_loss: 2.1888
Epoch 6/10
225/225              3s 15ms/step -
accuracy: 0.1390 - loss: 2.1898 - val_accuracy: 0.1094 - val_loss: 2.1888
Epoch 7/10
225/225              3s 15ms/step -
accuracy: 0.1391 - loss: 2.1899 - val_accuracy: 0.1094 - val_loss: 2.1889
Epoch 8/10
225/225              4s 16ms/step -
accuracy: 0.1391 - loss: 2.1899 - val_accuracy: 0.1441 - val_loss: 2.1889
Epoch 9/10
225/225              4s 16ms/step -
accuracy: 0.1397 - loss: 2.1899 - val_accuracy: 0.1441 - val_loss: 2.1889
Epoch 10/10
225/225              4s 16ms/step -
accuracy: 0.1394 - loss: 2.1900 - val_accuracy: 0.1441 - val_loss: 2.1889
```






```
100/100              0s 2ms/step
CNN Final Accuracy :0.144
CNN Duration(s) :34.54
CNN Final loss :2.189
CNN Final F1(Macro) :0.028
```

[23]: (0.144, 34.54, 2.189, 0.028)

1.4 3. Hyperparameter tuning

1.4.1 Algorithm of choice from first six weeks of course

[24]:
```
# RandomForest in hyperparamete tuning.

def bestRFClassifier():

    res = pd.DataFrame()

    n_estimators = [10, 20, 30]
    max_leaf_nodes = [10, 40, 70, 100]
    param_grid = {'n_estimators': n_estimators,
              'max_leaf_nodes': max_leaf_nodes}
```

```
    for paramater in ParameterGrid(param_grid):

        start = time.time() # Record the start time.
        rf = RandomForestClassifier(**paramater, criterion='entropy',␣
↪max_features = 'sqrt', n_jobs = -1, random_state=0)
        scores = cross_val_score(rf, flat_X_train_PCA, y_train, cv=cvKFold,␣
↪n_jobs=-1, scoring='accuracy')
        scores_f1_macro = cross_val_score(rf, flat_X_train_PCA, y_train,␣
↪cv=cvKFold, n_jobs=-1, scoring='f1_macro')
        scores_f1_weighted = cross_val_score(rf, flat_X_train_PCA, y_train,␣
↪cv=cvKFold, n_jobs = -1, scoring='f1_weighted')
        end = time.time() # Record the end time

        time_use = round(end - start, 3)
        accuracy_score = round(scores.mean(), 3) * 100
        f1_macro_score = round(scores_f1_macro.mean(), 3)
        f1_weighted_score = round(scores_f1_weighted.mean(), 3)

        df = pd.DataFrame([[time_use,
                            accuracy_score,
                            f1_macro_score,
                            f1_weighted_score,
                            paramater['n_estimators'],
                            paramater['max_leaf_nodes']
                            ]])
        res = pd.concat([res,df])

    res.columns = ['Duration(s)',
                   'Accuracy(%)',
                   'F1(Macro)',
                   'F1(Weighted)',
                   'n_estimators',
                   'max_leaf_nodes'
                   ]

    return res
```

[25]:
```
df_Rf = bestRFClassifier()
```

[26]:
```
df_Rf.sort_values('Accuracy(%)', ascending=False).head(20)
```

[26]:

	Duration(s)	Accuracy(%)	F1(Macro)	F1(Weighted)	n_estimators	\
0	1002.598	58.3	0.518	0.535	30	
0	98.297	57.0	0.501	0.518	30	
0	73.708	56.6	0.502	0.518	20	
0	64.206	55.4	0.482	0.500	20	
0	80.390	54.9	0.463	0.482	30	

```
0       40.086         54.7      0.487         0.503            10
0       32.795         53.6      0.471         0.487            10
0       53.643         52.9      0.442         0.461            20
0       27.818         51.7      0.437         0.456            10
0       46.855         44.0      0.335         0.355            30
0       17.510         39.7      0.300         0.316            10
0       31.339         38.9      0.287         0.303            20

   max_leaf_nodes
0             100
0              70
0             100
0              70
0              40
0             100
0              70
0              40
0              40
0              10
0              10
0              10
```

So, consider the Accuracy and the duration.

We choose 'n_estimators = 10' and 'max_leaf_nodes = 100'.

1.4.2 Fully connected neural network

[27]:
```
# Re-write the Search method to catch the Running Duration.

class Time_GridSearch(keras_tuner.GridSearch):
    def run_trial(self, trial, *args, **kwargs):

        start = time.time()
        history = super().run_trial(trial, *args, **kwargs)
        end = time.time()

        duration = end-start
        trial.hyperparameters.values["Duration"] = duration # Here add a fake␣
 ↪hyperparameters to store the duration

        return history

class Time_RandomSearch(keras_tuner.RandomSearch):
    def run_trial(self, trial, *args, **kwargs):

        start = time.time()
        history = super().run_trial(trial, *args, **kwargs)
```

```
        end = time.time()

        duration = end-start
        trial.hyperparameters.values["Duration"] = duration # Here add a fake␣
↪hyperparameters to store the duration

        return history
```

[28]:
```
# Build the model for MLP. This is used for RandomResearch.
def build_model_MLP(hp):
    model = keras.models.Sequential()

    # Add the input layer
    model.add(keras.layers.Input(shape=[28,28,3]))
    model.add(keras.layers.Flatten())


    # Add 2 hidden layers, treating the number of hidden neurons
    # and the activation function in each as hyperparameters to tune over
    for i in range(1, 3):
        model.add(
            keras.layers.Dense(
                units=hp.Choice(f"Dense {i}", values=[100, 200]),
                activation=hp.Choice(f"Activation {i}", values=["relu",␣
↪"sigmoid", "tanh"])
            )
        )

    # Add the output layer for 9 class classification
    model.add(keras.layers.Dense(9, activation="softmax"))

    # Set up the learning rate values to be tuned over and define the model
    learning_rate = hp.Choice('Learning_rate', values=[0.1, 0.01, 0.001])

    model.compile(optimizer=keras.optimizers.Adam(learning_rate=learning_rate),
                  loss='sparse_categorical_crossentropy',
                  metrics=['accuracy'])

    return model
```

[29]:
```
# MLP in hyperparamete tuning. Here we have the 108 trials, we select 30 random␣
↪trials.

def RandomMLPClassifier():

    tuner = Time_RandomSearch(
        hypermodel=build_model_MLP,
```

```
        objective="val_accuracy",
        max_trials=30,
        executions_per_trial=1,
        overwrite=True,
        directory="keras_tuning_results",
        seed = 0
        )

    X_train_MLP, X_valid_MLP, y_train_MLP, y_valid_MLP =␣
↪train_test_split(X_train, y_train, train_size=0.9, random_state=0)

    tuner.search(X_train_MLP, y_train_MLP, epochs=10,␣
↪validation_data=(X_valid_MLP, y_valid_MLP))

    res = pd.DataFrame()

    for trial in tuner.oracle.trials.values():

        df = pd.DataFrame([round(trial.score,3) * 100,
                           round(trial.hyperparameters.values['Duration'],3),
                           trial.hyperparameters.values['Dense 1'],
                           trial.hyperparameters.values['Activation 1'],
                           trial.hyperparameters.values['Dense 2'],
                           trial.hyperparameters.values['Activation 2'],
                           trial.hyperparameters.values['Learning_rate']
                          ]).T
        df.index = [str(int(trial.trial_id)+1)]

        res = pd.concat([res,df])

    res.columns = (['Accuracy(%)', 'Duration(s)', 'Dense 1', 'Activation 1',␣
↪'Dense 2', 'Activation 2', 'Learning rate'])

    return res
```

[30]:
```
df_MLP_Random = RandomMLPClassifier()
```

```
Trial 30 Complete [00h 00m 18s]
val_accuracy: 0.14406250417232513

Best val_accuracy So Far: 0.6384375095367432
Total elapsed time: 00h 07m 34s
```

[31]:
```
df_MLP_Random.sort_values('Accuracy(%)', ascending=False).head(20)
```

[31]:
```
    Accuracy(%) Duration(s) Dense 1 Activation 1 Dense 2 Activation 2  \
12         63.8       11.77     100         relu     200         relu
```

```
25        63.5      16.474     200         relu     100      sigmoid
24        62.5      17.318     200         relu     200         relu
1         61.5       14.56     200         relu     100         relu
4         60.6      15.013     200      sigmoid     100         relu
10        60.5      17.064     200      sigmoid     100         tanh
18        59.8      17.662     200         tanh     100         tanh
26        59.4      11.436     100         tanh     200         relu
21        59.0      11.718     100         tanh     100      sigmoid
14        58.6      17.283     200         tanh     100      sigmoid
27        54.9      11.789     100      sigmoid     200      sigmoid
29        52.7      17.351     200      sigmoid     200         relu
9         50.3      11.728     100         tanh     200         relu
22        49.5      10.932     100         tanh     100         tanh
20        49.3      17.783     200         tanh     100         tanh
6         45.9      16.741     200         tanh     200         tanh
19        41.6      17.037     200         tanh     100         relu
3         40.1       9.551     100         relu     200      sigmoid
23        39.7      17.762     200         relu     200         relu
8         37.1      18.248     200         tanh     200      sigmoid

   Learning rate
12         0.001
25         0.001
24         0.001
1          0.001
4          0.001
10         0.001
18         0.001
26         0.001
21         0.001
14         0.001
27          0.01
29          0.01
9           0.01
22          0.01
20          0.01
6           0.01
19           0.1
3           0.01
23          0.01
8            0.1
```

Here we fix the hyperparameter 'learning rate = 0.001'. Do GridResearch to the remain hyperparameter.

[32]:
```
# Build the model for MLP.
def build_model_MLP_2(hp):
```

```
    model = keras.models.Sequential()

    # Add the input layer
    model.add(keras.layers.Input(shape=[28,28,3]))
    model.add(keras.layers.Flatten())


    # Add 2 hidden layers, treating the number of hidden neurons
    # and the activation function in each as hyperparameters to tune over
    for i in range(1, 3):
        model.add(
            keras.layers.Dense(
                units=hp.Choice(f"Dense {i}", values=[100, 200]),
                activation=hp.Choice(f"Activation {i}", values=["relu",␣
↪"sigmoid", "tanh"])
            )
        )

    # Add the output layer for 9 class classification
    model.add(keras.layers.Dense(9, activation="softmax"))

    # Set up the learning rate values to be tuned over and define the model
    learning_rate = 0.001
    model.compile(optimizer=keras.optimizers.Adam(learning_rate=learning_rate),
                  loss='sparse_categorical_crossentropy',
                  metrics=['accuracy'])

    return model
```

[33]:
```
# MLP in hyperparamete tuning. Here we have the 36 trials.

def GridMLPClassifier():

    tuner = Time_GridSearch(
        hypermodel=build_model_MLP_2,
        objective="val_accuracy",
        max_trials=None,
        executions_per_trial=1,
        overwrite=True,
        directory="keras_tuning_results",
        seed = 0
        )

    X_train_MLP, X_valid_MLP, y_train_MLP, y_valid_MLP =␣
↪train_test_split(X_train, y_train, train_size=0.9, random_state=0)
```

```
    tuner.search(X_train_MLP, y_train_MLP, epochs=10,␣
↪validation_data=(X_valid_MLP, y_valid_MLP))

    res = pd.DataFrame()

    for trial in tuner.oracle.trials.values():

        df = pd.DataFrame([round(trial.score,3) * 100,
                           round(trial.hyperparameters.values['Duration'],3),
                           trial.hyperparameters.values['Dense 1'],
                           trial.hyperparameters.values['Activation 1'],
                           trial.hyperparameters.values['Dense 2'],
                           trial.hyperparameters.values['Activation 2']
                          ]).T
        df.index = [str(int(trial.trial_id)+1)]

        res = pd.concat([res,df])

    res.columns = (['Accuracy(%)', 'Duration(s)', 'Dense 1', 'Activation 1',␣
↪'Dense 2', 'Activation 2'])

    return res
```

[34]:
```
df_MLP_Grid = GridMLPClassifier()
```

```
Trial 36 Complete [00h 00m 18s]
val_accuracy: 0.5975000262260437

Best val_accuracy So Far: 0.645312488079071
Total elapsed time: 00h 08m 24s
```

[35]:
```
df_MLP_Grid.sort_values('Accuracy(%)', ascending=False).head(20)
```

[35]:

	Accuracy(%)	Duration(s)	Dense 1	Activation 1	Dense 2	Activation 2
23	64.5	16.828	200	relu	200	sigmoid
22	64.2	15.723	200	relu	200	relu
20	63.9	15.772	200	relu	100	sigmoid
2	63.8	11.767	100	relu	100	sigmoid
5	63.8	12.297	100	relu	200	sigmoid
6	63.3	12.927	100	relu	200	tanh
1	63.1	11.801	100	relu	100	relu
4	63.0	11.508	100	relu	200	relu
19	62.9	17.138	200	relu	100	relu
3	62.2	11.411	100	relu	100	tanh
27	61.3	14.858	200	sigmoid	100	tanh
10	61.1	11.799	100	sigmoid	200	relu
29	60.9	17.134	200	sigmoid	200	sigmoid

26	60.9	15.682	200	sigmoid	100	sigmoid
24	60.9	16.047	200	relu	200	tanh
21	60.9	15.707	200	relu	100	tanh
28	60.8	16.978	200	sigmoid	200	relu
25	60.6	14.741	200	sigmoid	100	relu
8	60.6	11.177	100	sigmoid	100	sigmoid
9	60.5	11.479	100	sigmoid	100	tanh

So, consider the Accuracy and the duration.

We choose 'Dense 1 = 100', 'Activation 1 = 'relu'', 'Dense 2 = 200', 'Activation 1 = 'tanh'', 'learning rate = 0.001'.

1.4.3 Convolutional neural network

[36]:
```
# Build the model for CNN.
def build_model_CNN(hp):
    model = keras.models.Sequential()

    # Add the input layer
    model.add(keras.layers.Input(shape=[28,28,3]))

    # Add 2 hidden layers, treating the number of hidden neurons
    # and the activation function in each as hyperparameters to tune over


    model.add(keras.layers.Conv2D(32, kernel_size=(3, 3), activation=hp.
 ↪Choice(f"Activation {1}", values=["relu", "sigmoid", "tanh"])),
              keras.layers.MaxPooling2D(pool_size=(2, 2)))

    model.add(keras.layers.Conv2D(64, kernel_size=(3, 3), activation=hp.
 ↪Choice(f"Activation {2}", values=["relu", "sigmoid", "tanh"])),
              keras.layers.MaxPooling2D(pool_size=(2, 2)))

    model.add(keras.layers.Flatten())
    Dropout_rate = hp.Choice('Dropout_rate', values=[0.2, 0.5, 0.8])
    model.add(keras.layers.Dropout(Dropout_rate))

    # Add the output layer for 9 class classification
    model.add(keras.layers.Dense(9, activation="softmax"))

    # Set up the learning rate values to be tuned over and define the model
    learning_rate = hp.Choice('Learning_rate', values=[0.1, 0.01, 0.001])
    opt = keras.optimizers.Adam(learning_rate=learning_rate)
    model.compile(loss='sparse_categorical_crossentropy',
                  optimizer=opt,
                  metrics=['accuracy'])
```

```
    return model
```

[37]:
```
# CNN in hyperparamete tuning. Here we have 81 trials, we choose random 20␣
↪trials.

def RandomCNNClassifier():

    tuner = Time_RandomSearch(
        hypermodel=build_model_CNN,
        objective="val_accuracy",
        max_trials=20,
        executions_per_trial=1,
        overwrite=True,
        directory="keras_tuning_results",
        seed = 0
        )

    X_train_MLP, X_valid_MLP, y_train_MLP, y_valid_MLP =␣
↪train_test_split(X_train, y_train, train_size=0.9, random_state=0)

    tuner.search(X_train_MLP, y_train_MLP, batch_size = 128, epochs=10,␣
↪validation_data=(X_valid_MLP, y_valid_MLP))

    res = pd.DataFrame()

    for trial in tuner.oracle.trials.values():

        df = pd.DataFrame([round(trial.score,3)*100,
                           round(trial.hyperparameters.values['Duration'],3),
                           trial.hyperparameters.values['Activation 1'],
                           trial.hyperparameters.values['Activation 2'],
                           trial.hyperparameters.values['Dropout_rate'],
                           trial.hyperparameters.values['Learning_rate'],
                          ]).T
        df.index = [str(int(trial.trial_id)+1)]

        res = pd.concat([res,df])

    res.columns = (['Accuracy(%)', 'Duration(s)', 'Activation 1', 'Activation␣
↪2', 'Dropout rate', 'Learning rate'])

    return res
```

[38]:
```
df_CNN_Random = RandomCNNClassifier()
```

```
Trial 20 Complete [00h 02m 20s]
val_accuracy: 0.7240625023841858
```

```
Best val_accuracy So Far: 0.8303124904632568
Total elapsed time: 00h 45m 23s
```

[39]:
```python
df_CNN_Random.sort_values('Accuracy(%)', ascending=False).head(20)
```

[39]:

	Accuracy(%)	Duration(s)	Activation 1	Activation 2	Dropout rate	Learning rate
17	83.0	135.613	relu	relu	0.5	0.001
19	78.1	136.884	relu	tanh	0.8	0.001
12	77.2	135.982	tanh	relu	0.2	0.001
1	76.8	134.683	relu	sigmoid	0.5	0.001
18	76.8	136.784	relu	tanh	0.8	0.001
16	73.5	140.827	tanh	sigmoid	0.5	0.001
20	72.4	140.0	tanh	sigmoid	0.2	0.001
15	69.3	139.485	tanh	sigmoid	0.2	0.01
11	67.8	135.633	relu	tanh	0.2	0.01
9	67.5	139.257	sigmoid	sigmoid	0.5	0.001
14	67.4	135.393	relu	tanh	0.5	0.01
5	64.8	135.495	sigmoid	relu	0.8	0.001
2	62.7	135.398	tanh	tanh	0.2	0.1
10	60.3	135.39	sigmoid	relu	0.8	0.001
7	57.8	134.833	relu	tanh	0.2	0.1
6	54.1	134.511	relu	tanh	0.8	0.1
13	49.6	134.727	tanh	relu	0.5	0.01
3	15.1	132.86	relu	relu	0.2	0.1
8	14.7	134.262	sigmoid	relu	0.5	
4	14.4	134.486	sigmoid	relu	0.2	

```
8            0.1
4           0.01
```

Here we fix the hyperparameter 'learning rate = 0.001' and 'activation 1 = relu'. Do GridResearch to the remain hyperparameter.

[40]:
```
def build_model_CNN_2(hp):
    model = keras.models.Sequential()

    # Add the input layer
    model.add(keras.layers.Input(shape=[28,28,3]))

    # Add 2 hidden layers, treating the number of hidden neurons
    # and the activation function in each as hyperparameters to tune over
    model.add(keras.layers.Conv2D(32, kernel_size=(3, 3),␣
 ↪activation='relu'),keras.layers.MaxPooling2D(pool_size=(2, 2)))

    model.add(keras.layers.Conv2D(64, kernel_size=(3, 3), activation=hp.
 ↪Choice(f"Activation {2}", values=["relu", "sigmoid", "tanh"])),
              keras.layers.MaxPooling2D(pool_size=(2, 2)))

    model.add(keras.layers.Flatten())
    Dropout_rate = hp.Choice('Dropout_rate', values=[0.2, 0.5, 0.8])
    model.add(keras.layers.Dropout(Dropout_rate))

    # Add the output layer for 9 class classification
    model.add(keras.layers.Dense(9, activation="softmax"))

    # Set up the learning rate values to be tuned over and define the model
    learning_rate = 0.001
    opt = keras.optimizers.Adam(learning_rate=learning_rate)
    model.compile(loss='sparse_categorical_crossentropy',
                  optimizer=opt,
                  metrics=['accuracy'])

    return model
```

[41]:
```
# CNN in hyperparamete tuning. Here we have grid search.

def GridCNNClassifier():

    tuner = Time_GridSearch(
        hypermodel=build_model_CNN_2,
        objective="val_accuracy",
        max_trials=None,
        executions_per_trial=1,
        overwrite=True,
```

```
        directory="keras_tuning_results",
        seed = 0
        )

    X_train_MLP, X_valid_MLP, y_train_MLP, y_valid_MLP =␣
↪train_test_split(X_train, y_train, train_size=0.9, random_state=0)

    tuner.search(X_train_MLP, y_train_MLP, batch_size = 128, epochs=10,␣
↪validation_data=(X_valid_MLP, y_valid_MLP))

    res = pd.DataFrame()

    for trial in tuner.oracle.trials.values():

        df = pd.DataFrame([round(trial.score,3)*100,
                           round(trial.hyperparameters.values['Duration'],3),
                           trial.hyperparameters.values['Activation 2'],
                           trial.hyperparameters.values['Dropout_rate']
                          ]).T
        df.index = [str(int(trial.trial_id)+1)]

        res = pd.concat([res,df])

    res.columns = (['Accuracy(%)', 'Duration(s)', 'Activation 2', 'Dropout␣
↪rate'])

    return res
```

[42]:
```
df_CNN_Grid = GridCNNClassifier()
```

```
Trial 9 Complete [00h 02m 15s]
val_accuracy: 0.7740625143051147

Best val_accuracy So Far: 0.8471875190734863
Total elapsed time: 00h 20m 30s
```

[43]:
```
df_CNN_Grid.sort_values('Accuracy(%)', ascending=False).head(20)
```

[43]:

	Accuracy(%)	Duration(s)	Activation 2	Dropout rate
3	84.7	135.127	relu	0.8
2	82.0	135.212	relu	0.5
1	80.1	135.619	relu	0.2
4	78.2	139.051	sigmoid	0.2
6	77.5	138.22	sigmoid	0.8
5	77.4	138.787	sigmoid	0.5
9	77.4	134.78	tanh	0.8
8	74.4	136.404	tanh	0.5

```
7        71.4      137.07         tanh          0.2
```

So, consider the Accuracy and the duration.

We choose 'Activation 1 = 'relu'', 'Activation 2 = 'relu'', 'Dropout rate = 0.8', 'learning rate = 0.001'.

[44]:
```
# Store the data to the csv document.
df_Rf.sort_values('Accuracy(%)', ascending=False, inplace=True)
df_MLP_Random.sort_values('Accuracy(%)', ascending=False, inplace = True)
df_MLP_Grid.sort_values('Accuracy(%)', ascending=False, inplace=True)
df_CNN_Random.sort_values('Accuracy(%)', ascending=False, inplace=True)
df_CNN_Grid.sort_values('Accuracy(%)', ascending=False, inplace=True)
df_Rf.to_csv(f'RF_{datetime.now()}.csv')
df_MLP_Random.to_csv(f'MLP_R{datetime.now()}.csv')
df_MLP_Grid.to_csv(f'MLP_G{datetime.now()}.csv')
df_CNN_Random.to_csv(f'CNN_R{datetime.now()}.csv')
df_CNN_Grid.to_csv(f'CNN_G{datetime.now()}.csv')
```

1.5 4. Final models

1.5.1 Algorithm of choice from first six weeks of course

For Random Forest model, we use hyperparameter list:

n_estimators = 10

max_leaf_nodes = 100

[45]:
```
acc_rf, Duration_rf, f1_m_rf, f1_w_rf = rf(n_estimators = 10,
    max_leaf_nodes = 100,
    X_test = flat_X_test_PCA,
    y_test = y_test
    )
```

```
Rf Accuracy: 0.546
Rf Duration(s): 4.686
Rf F1 Macro: 0.489
Rf F1 Weighted(s): 0.504
```

1.5.2 Fully connected neural network

For MLP model, we use hyperparameter list:

Dense_1 = 100

Activation_1 = 'relu'

Dense_2 = 200

Activation_2 = 'tanh'

Learning_rate = 0.001,

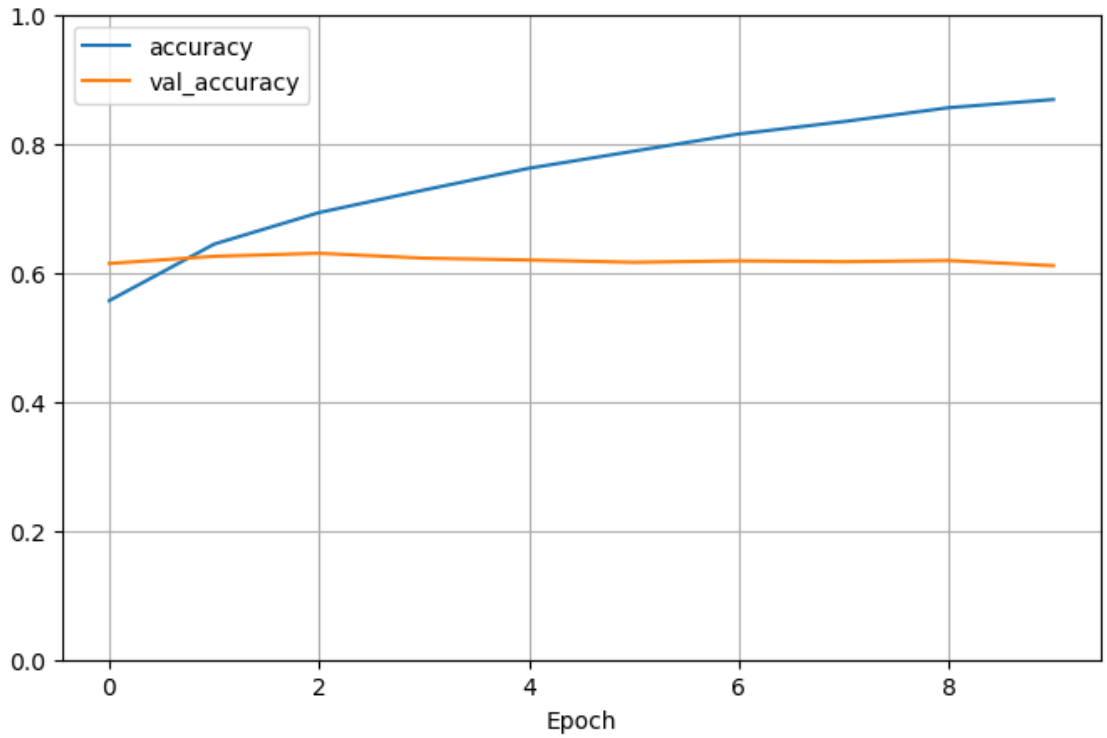
[46]:
```
acc_MLP,Duration_MLP,loss_MLP,f1_m_MLP = MLP(Dense_1 = 100,
        Activation_1 = 'relu',
        Dense_2 = 200,
        Activation_2 = 'tanh',
        Learning_rate = 0.001,
        X_test = X_test,
        y_test = y_test
        )
```

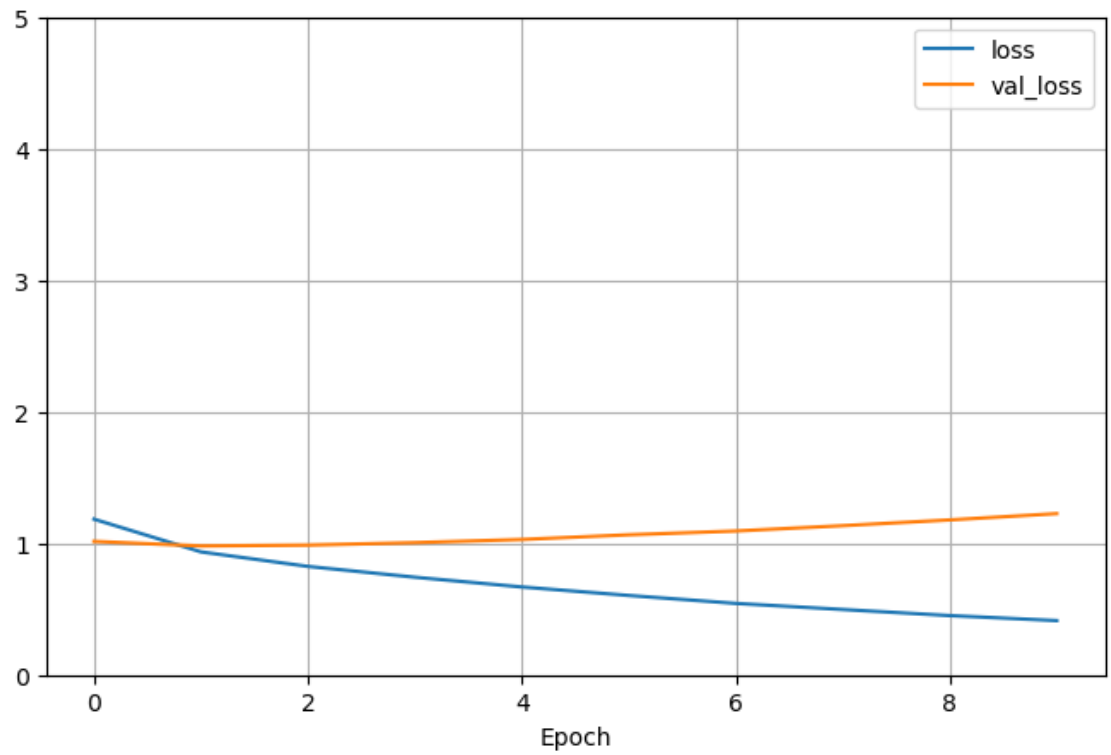
```
Model: "sequential"
```

Layer (type)	Output Shape	Param #
flatten (Flatten)	(None, 2352)	0
dense (Dense)	(None, 100)	235,300
dense_1 (Dense)	(None, 200)	20,200
dense_2 (Dense)	(None, 9)	1,809

```
Total params: 257,309 (1005.11 KB)

Trainable params: 257,309 (1005.11 KB)

Non-trainable params: 0 (0.00 B)

Epoch 1/10
225/225              1s 2ms/step -
accuracy: 0.4930 - loss: 1.3757 - val_accuracy: 0.6143 - val_loss: 1.0152
Epoch 2/10
225/225              0s 2ms/step -
accuracy: 0.6316 - loss: 0.9640 - val_accuracy: 0.6254 - val_loss: 0.9806
Epoch 3/10
225/225              1s 3ms/step -
accuracy: 0.6854 - loss: 0.8412 - val_accuracy: 0.6301 - val_loss: 0.9872
Epoch 4/10
225/225              1s 4ms/step -
accuracy: 0.7228 - loss: 0.7558 - val_accuracy: 0.6225 - val_loss: 1.0058
Epoch 5/10
225/225              1s 5ms/step -
accuracy: 0.7570 - loss: 0.6802 - val_accuracy: 0.6196 - val_loss: 1.0298
Epoch 6/10
```

```
225/225              1s 3ms/step -
accuracy: 0.7880 - loss: 0.6119 - val_accuracy: 0.6161 - val_loss: 1.0655
Epoch 7/10
225/225              1s 3ms/step -
accuracy: 0.8125 - loss: 0.5514 - val_accuracy: 0.6184 - val_loss: 1.0941
Epoch 8/10
225/225              1s 2ms/step -
accuracy: 0.8325 - loss: 0.5019 - val_accuracy: 0.6170 - val_loss: 1.1348
Epoch 9/10
225/225              1s 2ms/step -
accuracy: 0.8540 - loss: 0.4554 - val_accuracy: 0.6189 - val_loss: 1.1785
Epoch 10/10
225/225              1s 2ms/step -
accuracy: 0.8674 - loss: 0.4188 - val_accuracy: 0.6109 - val_loss: 1.2263
```






```
250/250              0s 643us/step
MLP Final Accuracy :0.611
MLP Duration(s) :6.959
MLP Final Loss :1.226
MLP Final F1(Macro) :0.587
```

1.5.3 Convolutional neural network

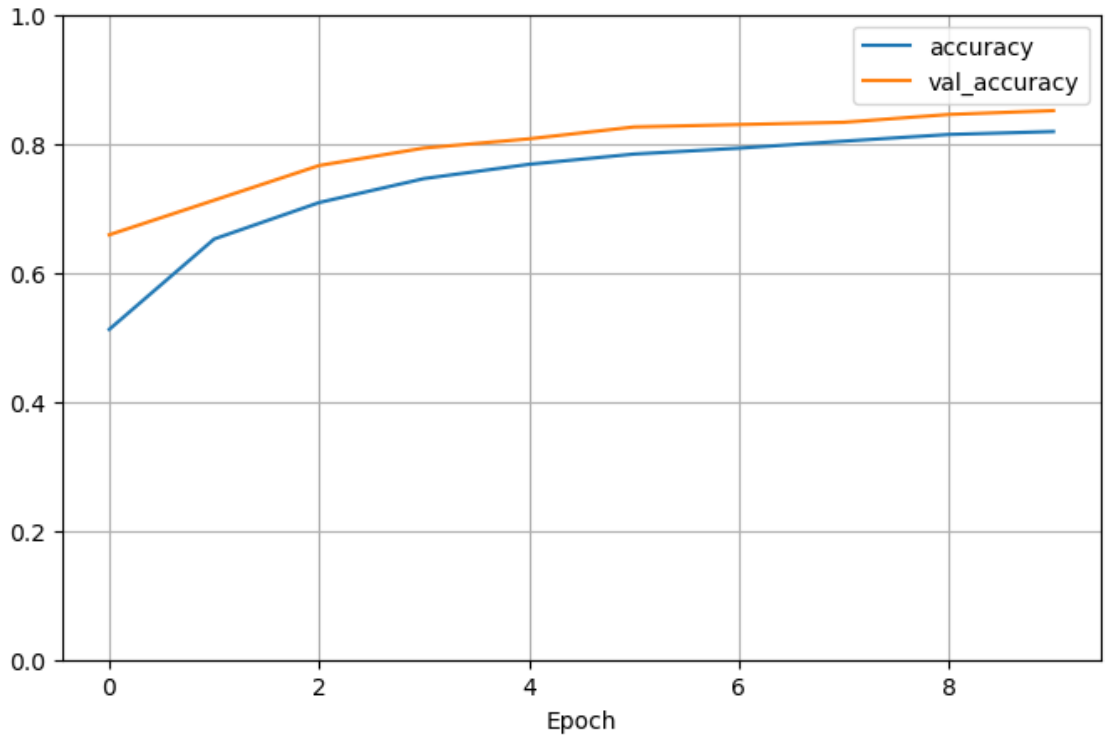
[47]:
```
acc_CNN,Duration_CNN,loss_CNN,f1_m_CNN = CNN(Activation_1 = 'relu',
        Activation_2 = 'relu',
        Dropout_rate = 0.8,
        Learning_rate = 0.001,
        X_test = X_test,
        y_test = y_test
        )
```

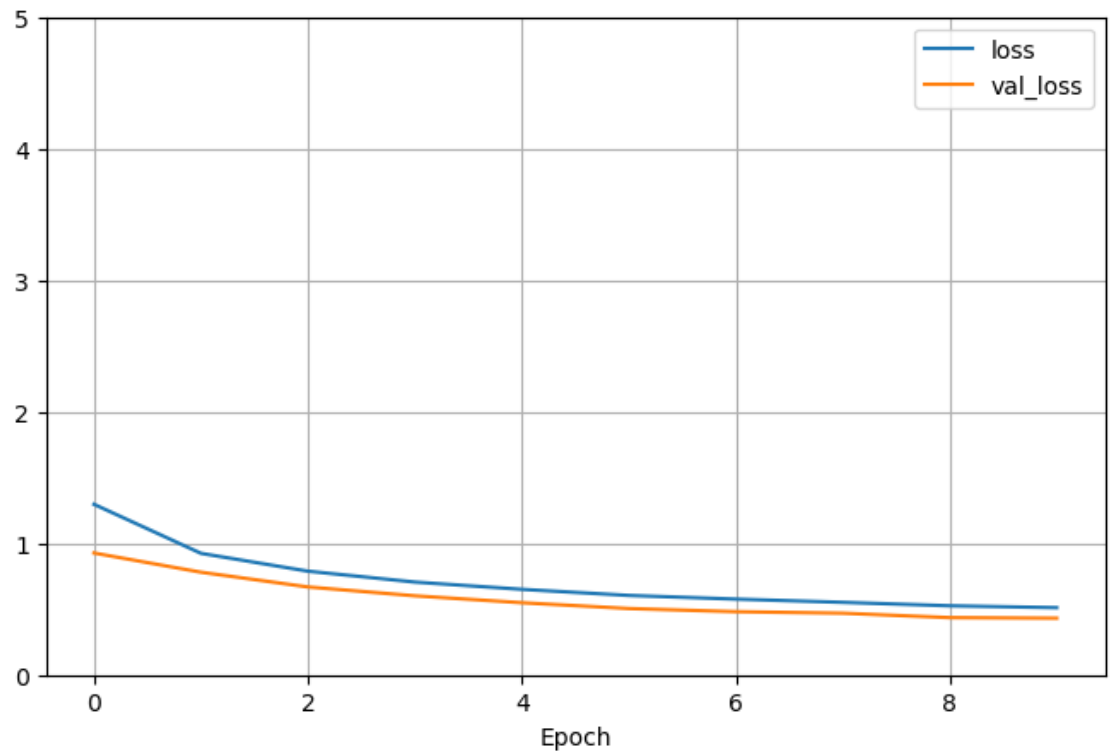
```
Model: "sequential_1"
```

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 26, 26, 32)	896

```
 max_pooling2d (MaxPooling2D)      (None, 13, 13, 32)                  0

 conv2d_1 (Conv2D)                 (None, 11, 11, 64)             18,496

 max_pooling2d_1 (MaxPooling2D)    (None, 5, 5, 64)                    0

 flatten_1 (Flatten)               (None, 1600)                        0

 dropout (Dropout)                 (None, 1600)                        0

 dense_3 (Dense)                   (None, 9)                      14,409


 Total params: 33,801 (132.04 KB)

 Trainable params: 33,801 (132.04 KB)

 Non-trainable params: 0 (0.00 B)

Epoch 1/10
225/225              5s 21ms/step -
accuracy: 0.4093 - loss: 1.5954 - val_accuracy: 0.6587 - val_loss: 0.9269
Epoch 2/10
225/225              4s 19ms/step -
accuracy: 0.6350 - loss: 0.9679 - val_accuracy: 0.7124 - val_loss: 0.7809
Epoch 3/10
225/225              4s 19ms/step -
accuracy: 0.6977 - loss: 0.8186 - val_accuracy: 0.7661 - val_loss: 0.6689
Epoch 4/10
225/225              4s 19ms/step -
accuracy: 0.7393 - loss: 0.7228 - val_accuracy: 0.7933 - val_loss: 0.6016
Epoch 5/10
225/225              4s 19ms/step -
accuracy: 0.7633 - loss: 0.6620 - val_accuracy: 0.8075 - val_loss: 0.5490
Epoch 6/10
225/225              4s 19ms/step -
accuracy: 0.7834 - loss: 0.6113 - val_accuracy: 0.8259 - val_loss: 0.5049
Epoch 7/10
225/225              5s 20ms/step -
accuracy: 0.7899 - loss: 0.5835 - val_accuracy: 0.8296 - val_loss: 0.4809
Epoch 8/10
225/225              4s 20ms/step -
accuracy: 0.8030 - loss: 0.5554 - val_accuracy: 0.8331 - val_loss: 0.4694
Epoch 9/10
225/225              4s 20ms/step -
accuracy: 0.8132 - loss: 0.5304 - val_accuracy: 0.8453 - val_loss: 0.4357
```

```
Epoch 10/10
225/225              5s 20ms/step -
accuracy: 0.8193 - loss: 0.5096 - val_accuracy: 0.8512 - val_loss: 0.4315
```






```
250/250              1s 2ms/step
CNN Final Accuracy :0.851
CNN Duration(s) :44.719
CNN Final loss :0.431
CNN Final F1(Macro) :0.844
```

The final result is shown:

[48]:
```
print(f'Final RandomForest Accuracy on test set: {acc_rf}')
print(f'Final RandomForest Duration on test set: {Duration_rf}')
print(f'Final RandomForest F1(Macro) on test set: {f1_m_rf}')
print(f'Final RandomForest F1(Weighted) on test set: {f1_w_rf}')
print()
print(f'Final MLP Accuracy on test set: {acc_MLP}')
print(f'Final MLP Duration on test set: {Duration_MLP}')
print(f'Final MLP Loss on test set: {loss_MLP}')
print(f'Final MLP F1(Macro) on test set: {f1_m_MLP}')
print()
print(f'Final CNN Accuracy on test set: {acc_CNN}')
print(f'Final CNN Duration on test set: {Duration_CNN}')
print(f'Final CNN Loss on test set: {loss_CNN}')
print(f'Final CNN F1(Macro) on test set: {f1_m_CNN}')
```

```
Final RandomForest Accuracy on test set: 0.546
```

```
Final RandomForest Duration on test set: 4.686
Final RandomForest F1(Macro) on test set: 0.489
Final RandomForest F1(Weighted) on test set: 0.504

Final MLP Accuracy on test set: 0.611
Final MLP Duration on test set: 6.959
Final MLP Loss on test set: 1.226
Final MLP F1(Macro) on test set: 0.587

Final CNN Accuracy on test set: 0.851
Final CNN Duration on test set: 44.719
Final CNN Loss on test set: 0.431
Final CNN F1(Macro) on test set: 0.844
```